MLDL Practical 8

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Aim: Design and train a Convolutional Neural Network (CNN) on image datasets

Dataset Source

Dataset Name: Fashion-MNIST Dataset

Platform: TensorFlow Keras (built-in)

Dataset Link: tensorflow.keras.datasets.fashion_mnist

Dataset Description

The Fashion-MNIST dataset is a multi-class image classification dataset consisting of grayscale images of 10 clothing categories. It serves as a more challenging replacement for the original MNIST digit dataset due to inter-class visual similarity.

Dataset Characteristics

- **Number of instances:** 70,000 images (60,000 training, 10,000 testing)
- **Image Dimensions:** 28 × 28 pixels, grayscale (1 channel)
- **Target Variable:** 10 clothing categories — T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag, Ankle boot
- **Normalization:** Pixel values scaled from [0, 255] to [0, 1]

Since pixel values span a wide numerical range, normalization is essential for stable gradient descent and faster CNN convergence.

Mathematical Formulation

A Convolutional Neural Network consists of Convolutional Layers, Pooling Layers, Fully Connected (Dense) Layers, and an Output Layer.

Convolution Operation

The core of a CNN is the discrete convolution, where a kernel (filter) K slides over image I:

$$(I * K)(i, j) = \Sigma m \, \Sigma n \, I(i+m, j+n) \cdot K(m, n)$$

This allows the network to learn local spatial patterns such as edges and textures through shared weights.

Activation and Downsampling

- **ReLU:** Applied after each convolution to introduce non-linearity: $f(x) = \max(0, x)$
- **Max Pooling:** Reduces spatial dimensions by selecting the maximum value in a 2×2 window, reducing computation while providing translation invariance.

Loss Function – Sparse Categorical Cross-Entropy

Since Fashion-MNIST is a 10-class problem with integer labels, Sparse Categorical Cross-Entropy is used:

$$L = -(1/N) \sum_i \log(\hat{y}_i[y_i])$$

Where y_i is the true class label and $\hat{y}_i[y_i]$ is the predicted probability for that class. This loss is minimized via backpropagation and the Adam optimizer.

Algorithm Limitations

- **Computational Intensity:** Training CNNs requires significantly more GPU power and memory compared to standard ANNs.
- **Data Requirement:** CNNs are prone to overfitting without sufficient data volume or data augmentation.
- **Resolution Sensitivity:** CNNs struggle with images of varying sizes unless uniformly resized at input.
- **Black-box Nature:** Learned convolutional filters are difficult to interpret directly.

Methodology / Workflow

Step	Phase	Action
1	Data Acquisition	Load Fashion-MNIST dataset directly via tensorflow.keras.datasets
2	Preprocessing	Normalize pixel values to [0, 1] and reshape images to (28, 28, 1)
3	Feature Extraction	Stack Conv2D layers with ReLU activation for spatial pattern recognition
4	Downsampling	Apply MaxPooling2D layers to reduce spatial dimensions
5	Regularization	Add Dropout(0.3) layer to prevent overfitting
6	Classification	Flatten feature maps and pass through Dense layers with Softmax output
7	Compilation	Use Adam optimizer with Sparse Categorical Cross-Entropy loss
8	Evaluation	Plot Accuracy/Loss curves and generate Confusion Matrix

Hyperparameter Tuning

The following hyperparameters were selected to optimize model performance on Fashion-MNIST:

Hyperparameter	Value	Rationale
Conv Filters (Layer 1)	32	Extracts low-level features like edges from 28×28 input
Conv Filters (Layers 2 & 3)	64	Captures higher-level spatial patterns and textures
Kernel Size	(3, 3)	Standard receptive field balancing detail and computation
Pooling Size	(2, 2)	Halves spatial dimensions without losing structural information

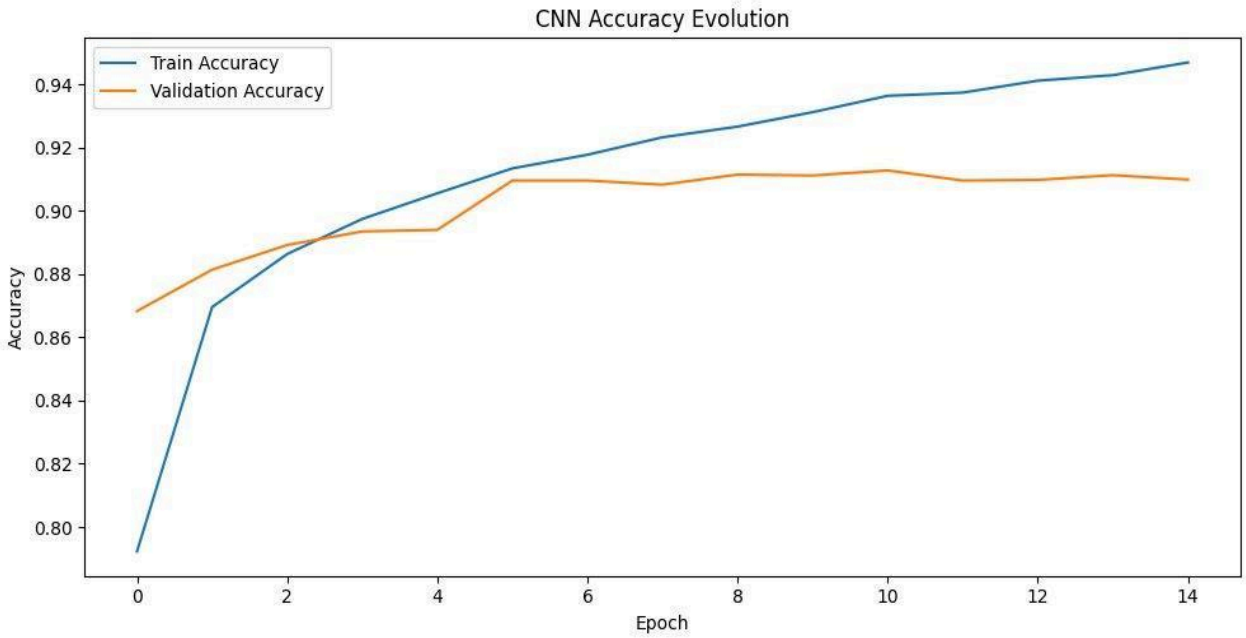
Dense Units	64	Sufficient capacity for 10-class separation after flattening
Dropout Rate	0.3	Moderate regularization reducing overfitting without underfitting
Optimizer	Adam	Adaptive learning rate for efficient and stable convergence
Learning Rate	0.001	Standard Adam default ensuring smooth weight updates
Epochs	15	Sufficient for accuracy to plateau without excessive overfitting
Batch Size	32	Balances gradient noise with computational efficiency

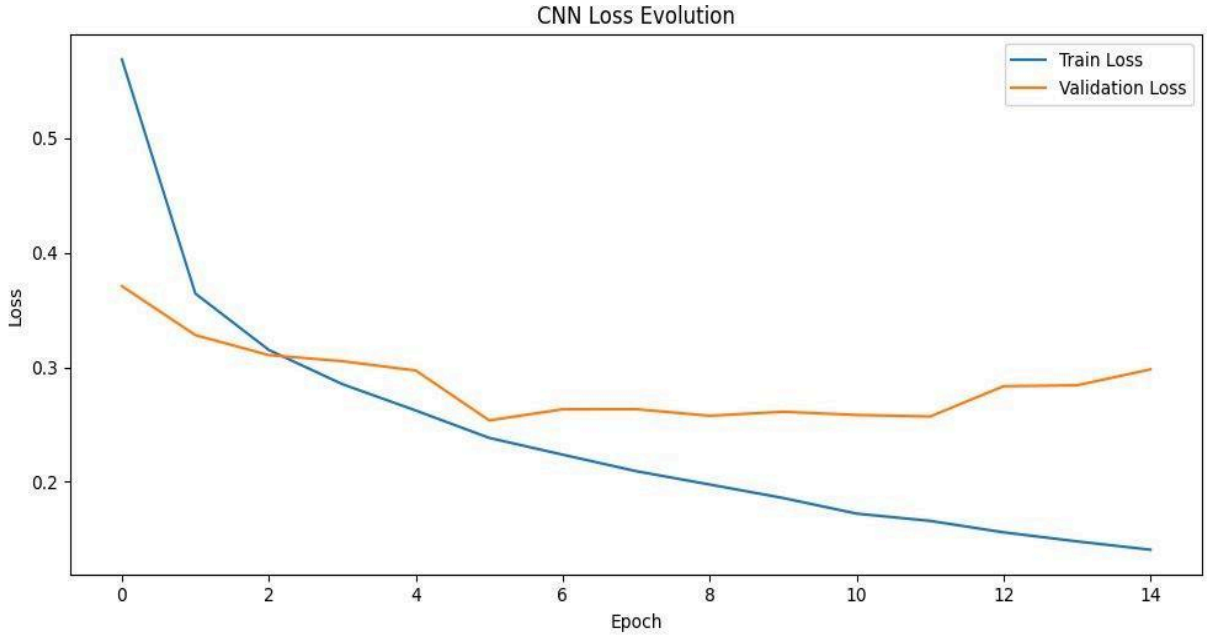
Performance Analysis

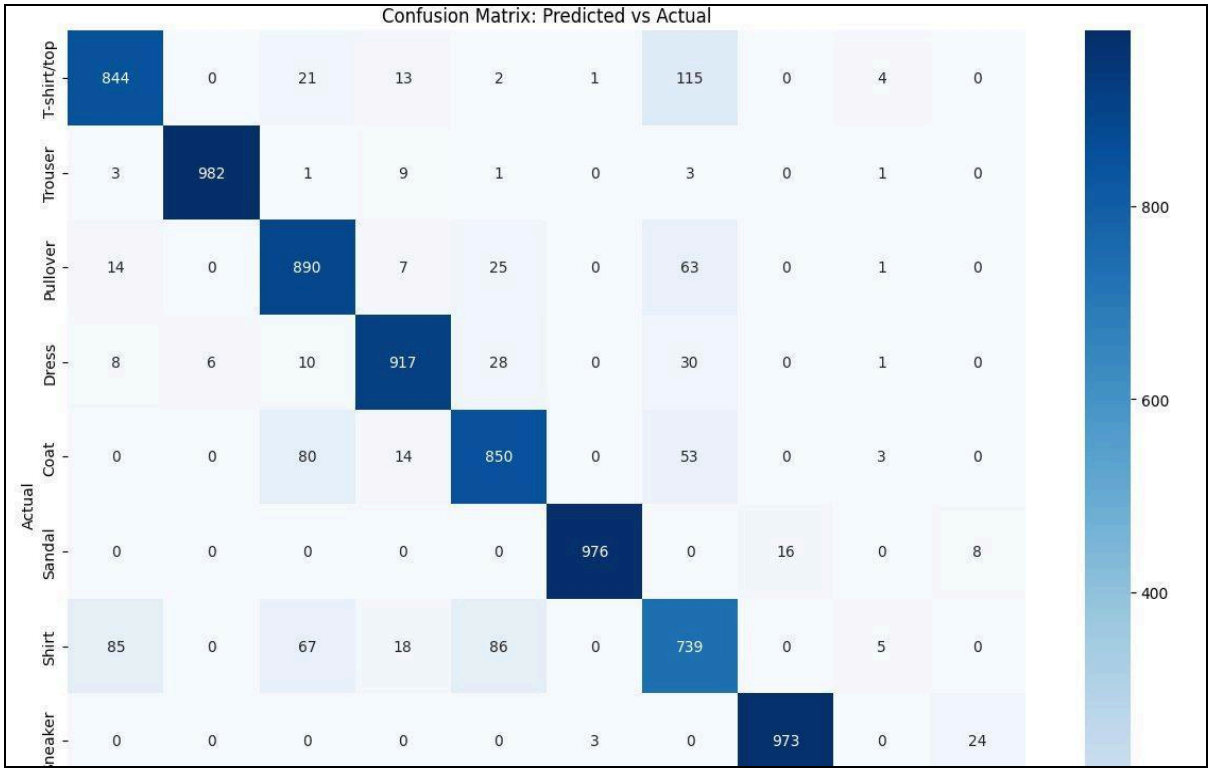
- The CNN achieved a final test accuracy of 90.99%, significantly outperforming traditional fully connected networks on image data.
- Dropout regularization narrowed the training-validation accuracy gap, improving generalization across unseen clothing images.
- The confusion matrix reveals that Shirt is the most misclassified category, frequently confused with T-shirt/top, Pullover, and Coat due to structural similarity.
- Classes such as Trouser, Sandal, and Sneaker achieved near-perfect recall (98–99%), as they are visually distinct from other categories.
- Validation accuracy plateaus near epoch 10–11 at ~91%, suggesting further improvement would require data augmentation or a deeper architecture.

Class	**Precision**	**Recall**	**F1-Score**	**Support**
T-shirt/top	0.88	0.84	0.86	1000
Trouser	0.99	0.98	0.99	1000
Pullover	0.83	0.89	0.86	1000
Dress	0.93	0.92	0.93	1000
Coat	0.86	0.85	0.85	1000
Sandal	0.99	0.98	0.98	1000
Shirt	0.73	0.74	0.73	1000
Sneaker	0.95	0.97	0.96	1000
Bag	0.98	0.97	0.98	1000
Ankle boot	0.96	0.96	0.96	1000
Accuracy			**0.91**	**10000**
Macro Avg	0.91	0.91	0.91	10000
Weighted Avg	0.91	0.91	0.91	10000

Output


CNN Accuracy Evolution
Train Accuracy
Validation Accuracy
Accuracy
Epoch
0.94
0.92
0.90
0.88
0.86
0.84
0.82
0.80
0
2
4
6
8
10
12
14


CNN Loss Evolution
Train Loss
Validation Loss
Loss
Epoch
0.5
0.4
0.3
0.2
0
2
4
6
8
10
12
14



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Final Test Accuracy: 90.99%
Final Test Loss: 0.2981

Conclusion

In this experiment, a Convolutional Neural Network was successfully implemented for clothing image classification using the Fashion-MNIST dataset.

Convolution layers, max pooling, dropout regularization, and the Adam optimizer played a crucial role in achieving stable training and strong generalization performance, with a final test accuracy of 90.99%.

This experiment demonstrates how CNNs effectively capture complex spatial hierarchies in image data and highlights their clear advantage over traditional fully connected networks for computer vision tasks.